NGUYEN DUY TRANG

MOBILE ENGINEER

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PROFESSIONAL SUMMARY

OVERVIEW

- **Core Mobile Engineering:** Over 5 years of production experience engineering high-performance mobile applications on Flutter, Swift (iOS), and Kotlin (Android), with deep native bridge capabilities.
- **Adaptability & Problem Solving:** Hands-on production track record across Travel OTT Social Networks, Fintech, Multi-vertical POS & Self-order Kiosks, and Multimodal AI integration. Pragmatically combines native platform power with Flutter to maximize performance and accelerate time-to-market.
- **Team Collaboration & Ownership:** Highly collaborative with cross-functional partners (Product, UI/UX, Backend, QA). Proactively surfaces architectural solutions and unblocks team velocity to deliver robust releases on schedule.

KEY HIGHLIGHTS & ACHIEVEMENTS

METRICS

- **Measured Performance Optimization:** Reduced self-order Kiosk checkout time by 30%; optimized camera buffer streaming to achieve sub-2.5s AI inference latency with locked 60 FPS rendering (0 memory leaks).
- **Offline-First — Local Database:** Embedded a high-performance Rust core into Flutter via Dart FFI to directly manage the Local Database (SQLite) and execute transactional business logic, guaranteeing zero-latency response and 100% offline persistence.
- **Automated Unit Testing & Self-Testing:** Committed to rigorous engineering self-testing: designed and maintained a comprehensive suite of 277 automated unit and widget tests (100% pass rate) covering core logic and state, integrated into CI/CD.
- **1M+ Production Scale & Reliability:** Proven track record maintaining large-scale consumer applications reaching 1M+ downloads across iOS and Android, sustaining a crash-free session rate above 99.5% in production.

TECHNICAL SKILLS MATRIX

COMPETENCIES

Languages & Platforms:	Flutter & Dart, Swift (iOS), Kotlin (Android), Rust, TypeScript.
Architecture & State:	Clean Architecture, Riverpod, BLoC, MVVM, VIPER, Modularization.
Networking & Data:	Offline-First, SQLite, Realm, Secure Storage, REST APIs (Dio), WebSocket, Firebase, Dart FFI.
Performance & Advanced:	Flutter DevTools, Render Tree Optimization, AutoRoute, Gemini AI, GenUI (A2UI).
Testing & CI/CD:	Unit & Widget Testing (277 Tests), GitHub Actions, GitLab CI/CD, Fastlane.
Delivery & Monitoring:	App Store Connect, Google Play Console, Firebase App Distribution, Sentry, Crashlytics, GA4.

COMMERCIAL WORK EXPERIENCE

TRACK RECORD

FINVIET TECHNOLOGY • Mobile Developer (Flutter, POS & Kiosk)

08/2025 – 08/2026

Tech Stack: Flutter, Dart, Clean Architecture, BLoC, Dio, SQLite/Drift, Platform Channels (Kotlin), Fastlane, CI/CD.

- Architected modular **Clean Architecture / MVVM** applying **BLoC (flutter_bloc)** with Unidirectional Data Flow for multi-vertical POS & self-order Kiosk systems (F&B, Retail, Spa), reducing transaction duration by **30%**.
- Engineered a centralized **Base UI Library & Responsive Layout** engine adapting dynamically across varied viewports (Mobile, Tablet, Kiosk/Web) with reusable custom widgets and themes.
- Constructed **Base Network Client with Dio**: automated token attachment interceptors, expired token refresh handling (401), exponential retry backoff, and offline cache policies.
- Designed **Offline-first data pipeline** (BLoC, SQLite/Drift), ensuring 100% order capture resilience during network outages with seamless background synchronization upon reconnection.
- Profiled render tree execution and eradicated drop frames/jank using **Flutter DevTools** (Profiler, Memory), eliminating memory leaks across ESC/POS thermal printers and hardware barcode scanner integrations.
- Optimized UI rendering and memory footprint for low-spec Kiosk hardware (Android POS terminal box, 10.1-inch secondary screen), locking steady **60 FPS**.
- Developed deep Native Platform Channels (Kotlin/Android) interfacing with peripheral hardware: ESC/POS thermal receipt printers, magnetic stripe / EMV payment terminals, and USB/Bluetooth barcode scanners.
- Established automated CI/CD build and delivery pipelines for Android and iOS using **Fastlane** and **GitHub Actions**.

HAHALOLO • Flutter Developer (Social Network & OTT Chat)

05/2021 – 07/2025

Tech Stack: Flutter, Dart, Riverpod, WebSocket, SQLite, Clean Architecture, Push Notifications, Media Cache.

- Engineered and maintained mission-critical features on a high-scale Travel Social Network & OTT Chat application serving **1M+ active users**.
- Implemented **Riverpod state management** under Clean Architecture patterns, enforcing rigorous boundaries between Presentation, Domain, and Data layers.
- Optimized multimedia feed rendering performance utilizing `ListView.builder`, `AutomaticKeepAliveClientMixin`, and localized disk caching, reducing runtime RAM consumption by **40%**.
- Engineered **Real-time WebSocket Chat subsystem**: bidirectional instant messaging, message deduplication and retry queues during disconnection, offline persistence with SQLite, and background push notifications (FCM / APNs).
- Integrated rich social features: personalized News Feed, multi-level comments, interactive reactions, post sharing, and travel geotagging.
- Optimized application cold start latency to **< 1.8 seconds**, maintaining a crash-free session rate **> 99.5%** in production.
- Authored comprehensive Unit and Widget Tests for core business flows; participated in rigorous PR code reviews prior to staging merges.

FEATURED PERSONAL PROJECT

FLAGSHIP ARCHITECTURE

ASTROBITE — AI Food Scanner & Nutrition Tracker

2026 - Present

Personal Project | Flutter, Dart, Riverpod, Firebase, Google Gemini AI

Tech Stack: Flutter, Dart, Riverpod, AutoRoute, Firebase Authentication, Cloud Firestore, Firebase App Check, Google Generative AI SDK (Gemini Vision & NLU), SharedPreferences, Open Food Facts API, Fastlane, GitHub Actions.

- **Designed and developed a cross-platform Flutter application** using Feature-First Architecture and Riverpod, implementing user authentication, food tracking, calorie and macronutrient dashboards, nutrition analytics, custom recipes, and meal planning.
- **Integrated Gemini multimodal Vision to analyze meal images** and extract structured nutritional data, including multi-dish decomposition, broth and topping estimation; added barcode lookup through Open Food Facts and AI-powered nutrition label parsing.
- **Implemented offline-capable food logging** with local caching, a pending-sync queue, duplicate handling, and Firestore-backed persistence and real-time data streams, supported by user-scoped Firestore security rules.
- **Built a context-aware AI Nutrition Coach** using Gemini multi-turn conversations, persisted chat history, and custom interactive Flutter components for in-chat meal logging.
- **Developed Vietnamese voice-based meal logging** by combining speech-to-text recognition with Gemini NLU to transform natural-language transcripts into structured nutrition records.
- **Automated Android and iOS release build workflows** using Fastlane and GitHub Actions, with successful release build jobs and artifact generation for both platforms.

EDUCATION

ACADEMIC

FPT POLYTECHNIC

2018 - 2021

Major: Software Engineering (Mobile Application Development)

LANGUAGES

PROFICIENCY

English: Reads and understands technical documentation, technical communication.